

Chapter 5

Applications of Differentiation

In this chapter, we will discuss applications of differentiation to curve sketching and extremal problems. For curve sketching, we need to consider geometric meanings of the first and second order derivatives. For convenience, some of the concepts and results are given not in their most general forms. Many of the concepts and results are stated for functions that are differentiable, twice differentiable etc. Below are the meanings of these terms.

Terminology Let f be a function that is defined on an open interval (a, b) . We say that

- f is *differentiable* on (a, b) if $f'(x)$ exists for all $x \in (a, b)$;
- f is *twice differentiable* on (a, b) if $f''(x)$ exists for all $x \in (a, b)$.

Explanation The condition “ f is a function defined on an open interval (a, b) ” means that $(a, b) \subseteq \text{dom}(f)$.

Terminology Let f be a function and let x_0 be a real number. We say that

- f is *defined on an open interval containing x_0* if there is an open interval (a, b) such that $x_0 \in (a, b)$ and $(a, b) \subseteq \text{dom}(f)$;
- f is *differentiable on an open interval containing x_0* if there is an open interval (a, b) such that $x_0 \in (a, b)$ and f is differentiable on (a, b) ;

Remark If f is a function defined on an open interval containing x_0 , then we may consider continuity and differentiability of f at x_0 .

Example Let $f(x) = \sqrt{x}$. The domain of f is $[0, \infty)$.

- Although $0 \in \text{dom}(f)$, the function f is not defined on an open interval containing 0. This is because there does not exist any open interval (a, b) such that $0 \in (a, b)$ and $(a, b) \subseteq \text{dom}(f)$.
- If x_0 is a positive real number, then f is defined on an open interval containing x_0 . This is because $x_0 \in (0, \infty)$ and $(0, \infty) \subseteq \text{dom}(f)$.

For $x > 0$, by the Power Rule, we have $f'(x) = \frac{1}{2\sqrt{x}}$. Thus, f is differentiable on $(0, \infty)$.

- If x_0 is a positive real number, then f is differentiable on an open interval containing x_0 .

For $x > 0$, by the Constant Multiple Rule and the Power Rule, we have $f''(x) = \frac{-1}{4\sqrt{x^3}}$. Thus, f is twice differentiable on $(0, \infty)$.

5.1 Curve Sketching

5.1.1 Increasing and Decreasing Functions

Definition Let f be a function and let I be an interval such that $I \subseteq \text{dom}(f)$. We say that f is

- *strictly increasing* on I if for any two numbers $x_1, x_2 \in I$, where $x_1 < x_2$, we have $f(x_1) < f(x_2)$;
- *strictly decreasing* on I if for any two numbers x_1, x_2 in I , where $x_1 < x_2$, we have $f(x_1) > f(x_2)$.

Remark

- In the definition, I can be an open interval, a closed interval or a half-open half-closed interval in the form $[a, b)$ or $(a, b]$.
- Although we can define the concepts “ f is strictly increasing (or strictly decreasing) on a set S , where $S \subseteq \text{dom}(f)$ ”, we will not use such concepts in this course because the concepts “strictly increasing (or strictly decreasing) on an interval” are good enough for our consideration; moreover, a function strictly increasing (or strictly decreasing) on a set S_1 and also on a set S_2 may not be strictly increasing (or strictly decreasing) on $S_1 \cup S_2$.

Geometric Meaning A function is strictly increasing (respectively strictly decreasing) on an interval I means that for $x \in I$, the graph of f goes up (respectively goes down) as x goes from left to right.

Terminology For simplicity, instead of saying “strictly increasing”, we will say “increasing” etc.

Remark Some authors have a different definition for “increasing”. In that case, “increasing” and “strictly increasing” refer to different concepts.

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = x^3$. Then f is increasing on $\mathbb{R}$.

Reason If $x_1 < x_2$, then $x_1^3 < x_2^3$.

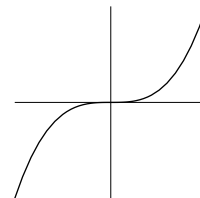


Figure 5.1

Example Let $f : (0, \infty) \rightarrow \mathbb{R}$ be the function given by $f(x) = \frac{1}{\sqrt{x}}$. Then f is decreasing on $(0, \infty)$.

Reason If $0 < x_1 < x_2$, then $\sqrt{x_1} < \sqrt{x_2}$ and so $\frac{1}{\sqrt{x_1}} > \frac{1}{\sqrt{x_2}}$.

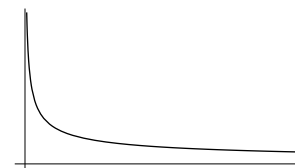


Figure 5.2

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = x^2$. Then f is decreasing on $(-\infty, 0)$ and increasing on $(0, \infty)$.

Reason

- If $x_1 < x_2 < 0$, then $x_1^2 > x_2^2$.
- If $0 < x_1 < x_2$, then $x_1^2 < x_2^2$.

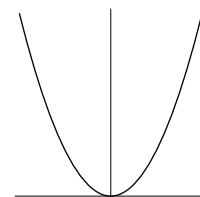


Figure 5.3

Remark

- If a function is increasing (respectively decreasing) on an interval I , then it is also increasing (respectively decreasing) on any interval J with $J \subseteq I$. For the function f in the above example, we can also say that f is increasing on $(0, 1]$; f is decreasing on $[-10, -2]$ etc.
- If a function is continuous on an interval $[a, b]$ and if it is increasing (respectively decreasing) on the open interval (a, b) , then it is increasing (respectively decreasing) on $[a, b]$. Similar results holds if $[a, b]$ is replaced by $(a, b]$ or $[a, b)$. For the function in the above example, it is decreasing on $(-\infty, 0]$ and increasing on $[0, \infty)$. These intervals are maximal in the sense that they cannot be enlarged.

Definition Let f be a function and let I be an interval with $I \subseteq \text{dom}(f)$ such that f is increasing (respectively decreasing) on I . We say that I is a *maximal interval* on which f is increasing (respectively decreasing) if there does not exist any interval J with $I \subsetneq J \subseteq \text{dom}(f)$ such that f is increasing (respectively decreasing) on J .

Example For the function f given in the preceding example, the interval $(-\infty, 0]$ is the maximal interval on which f is decreasing and $[0, \infty)$ is the maximal interval on which f is increasing.

In the above examples, we can determine where the function f is increasing or decreasing using inspection or using the graph of f . In general, given a function f , for example, $f(x) = 27x - x^3$, it is not easy to see where f is increasing or decreasing. For differentiable functions, the next theorem describes a simple way to determine where f is increasing or decreasing.

Theorem 5.1.1 Let f be a function that is defined and differentiable on an open interval (a, b) .

- (1) If $f'(x) > 0$ for all $x \in (a, b)$, then f is increasing on (a, b) .
- (2) If $f'(x) < 0$ for all $x \in (a, b)$, then f is decreasing on (a, b) .
- (3) If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b) , that is, $f(x_1) = f(x_2)$ for all $x_1, x_2 \in (a, b)$, or equivalently, there exists a real number c such that $f(x) = c$ for all $x \in (a, b)$.

Explanation In the first result, the condition “ $f'(x) > 0$ for all $x \in (a, b)$ ” means that the slope is always positive. From intuition, we “see” that the graph of f goes up. However, this is not a proof.

Proof The results can be proved rigorously using a result called the Mean-Value Theorem. For details, see Theorem B.3.1 in the Appendix. $\square$

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = 27x - x^3.$$

Find the interval(s), if any, on which f is increasing or decreasing.

Explanation

- The question is to find maximal interval(s) on which f is increasing and to find maximal interval(s) on which f is decreasing.

- Before getting the answer, we do not know whether there is any interval on which f is increasing or decreasing, so in the question, “if any” is added. Moreover, we do not know whether there are more than one intervals on which f is increasing or decreasing, so instead of asking for “interval”, we ask for “interval(s)”. This kind of wording is cumbersome; therefore, sometimes, we simply ask: “Find the intervals on which f is increasing or decreasing”.
- Because f is continuous, it suffices to find maximal open intervals on which f is increasing or decreasing. For this, we have to solve $f'(x) > 0$ or $f'(x) < 0$ respectively.

Solution Differentiating $f(x)$, we get

$$\begin{aligned} f'(x) &= \frac{d}{dx}(27x - x^3) \\ &= 27 - 3x^2 \\ &= 3(3 + x)(3 - x). \end{aligned}$$

	$(-\infty, -3)$	$(-3, 3)$	$(3, \infty)$
3	+	+	+
$3 + x$	−	+	+
$3 - x$	+	+	−
f'	−	+	−
f	$\searrow$	$\nearrow$	$\searrow$

From the table, we see that

- on the interval $[-3, 3]$, f is increasing;
- on the intervals $(-\infty, -3]$ and $[3, \infty)$, f is decreasing.

□

Remark

- In the last row of the above table, the first $\searrow$ indicates that f is decreasing on $(-\infty, -3)$ etc. This information is obtained from Theorem 5.1.1.
- Since f is decreasing on $(-\infty, -3)$, by continuity, it is decreasing on $(-\infty, 3]$ etc.

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x^4 - 4x^3 + 5.$$

Find where the function f is increasing or decreasing.

Explanation This example is similar to the last one. The question is to find maximal intervals on which f is increasing or decreasing.

Solution Differentiating $f(x)$, we get

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^4 - 4x^3 + 5) \\ &= 4x^3 - 12x^2 \\ &= 4x^2(x - 3) \end{aligned}$$

	$(-\infty, 0)$	$(0, 3)$	$(3, \infty)$
4	+	+	+
x^2	+	+	+
$x - 3$	−	−	+
f'	−	−	+
f	$\searrow$	$\searrow$	$\nearrow$

From the table, we see that

- on the interval $[3, \infty)$, f is increasing;
- on the interval $(-\infty, 3]$, f is decreasing.

□

Remark In the above example, to get the maximal interval on which f is decreasing, the following simple result is used.

Theorem 5.1.2 Let f be a function that is defined on an open interval (a, c) . If f is increasing (respectively decreasing) on (a, b) as well as on (b, c) and is continuous at b , then it is increasing (respectively decreasing) on (a, c) .

Proof We give the proof for the increasing case. Let $x_1, x_2 \in (a, c)$ and $x_1 < x_2$. We want to show that $f(x_1) < f(x_2)$. For this, we consider the following cases:

- (1) If $x_1, x_2 \in (a, b)$, then $f(x_1) < f(x_2)$ since f is increasing on (a, b) .
- (2) If $x_1, x_2 \in (b, c)$, then $f(x_1) < f(x_2)$ since f is increasing on (b, c) .
- (3) If $x_1 \in (a, b)$ and $x_2 = b$, then $f(x_1) < f(x_2)$ since f is increasing on (a, b) and continuous at b .
- (4) If $x_1 = b$ and $x_2 \in (b, c)$, then $f(x_1) < f(x_2)$ since f is increasing on (b, c) and continuous at b .
- (5) If $x_1 \in (a, b)$ and $x_2 \in (b, c)$, then $f(x_1) < f(b) < f(x_2)$ by Cases (3) and (4).

□

5.1.2 Relative Extrema

In the last section, we use Theorem 5.1.1 to find where a function f is increasing or decreasing. For that, we solve inequalities $f'(x) > 0$ and $f'(x) < 0$. In each of the tables obtained in the last two examples, the intervals are obtained from $\mathbb{R}$ by deleting the zeros of f' . For example, in the last example, $\mathbb{R} \setminus \{0, 3\} = (-\infty, 0) \cup (0, 3) \cup (3, \infty)$. Zeros of f' are important because they play an important role in extremum problems.

Definition Let f be a function and let x_0 be a real number such that f is defined on an open interval containing x_0 . If $f'(x_0) = 0$, then we say that x_0 is a *stationary number* of f .

Explanation

- If $x = t$ is time and $y = f(t)$ is the displacement (function) of a moving object, then $\frac{dy}{dt} = f'(t)$ is the velocity (function). Thus $f'(t_0) = 0$ means that the velocity at time t_0 is 0, that is, the object is stationary at that moment.
- In the definition, the condition “ f is defined on an open interval containing x_0 ” can be omitted because the condition “ $f'(x_0) = 0$ ” implicitly implies that f is defined on the left-side and right-side of x_0 as well as at x_0 . However, we will continue to use this lengthy description to give the general setting.

Definition Let f be a function and let x_0 be a real number such that f is defined on an open interval containing x_0 . If $f'(x_0)$ does not exist or $f'(x_0) = 0$, then we say that x_0 is a *critical number* of f .

Explanation

- Most functions considered in this course are differentiable (on open intervals that are subsets of their domains). For such functions, critical numbers and stationary numbers are the same.
- Instead of “critical number”, many authors use the term “critical point”. However, many students misunderstand this terminology and take $(x_0, f(x_0))$ as a critical point.

Caution A critical point is a point on the real line (that is, a real number) rather than a point on the graph of f (an ordered pair).

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = x^2 + 4x - 11$. Find the critical number(s) of f .

Explanation Because f is differentiable on $\mathbb{R}$, the question is to find the stationary numbers of f .

Solution Differentiating $f(x)$, we get
$$f'(x) = \frac{d}{dx}(x^2 + 4x - 11) = 2x + 4.$$

Solving $f'(x) = 0$, we get $x = -2$ which is the critical number of f . □

Definition Let f be a function and let x_0 be a real number such that f is defined on an open interval containing x_0 . We say that

- f has a *relative maximum* at $x = x_0$ if $f(x_0) \geq f(x)$ for all x sufficiently close to x_0 ;
- f has a *relative minimum* at $x = x_0$ if $f(x_0) \leq f(x)$ for all x sufficiently close to x_0 .

Remark The condition “ $f(x_0) \geq f(x)$ for all x sufficiently close to x_0 ” means that there exists an open interval (α, β) with $x_0 \in (\alpha, \beta)$ such that $f(x_0) \geq f(x)$ for all $x \in (\alpha, \beta)$. The interval (α, β) may be different from (a, b) .

Terminology Suppose that f has a relative maximum (respectively relative minimum) at x_0 . Then

- the number x_0 is called a *relative maximizer* (respectively *relative minimizer*) of f ;
- the ordered pair $(x_0, f(x_0))$ is called a *relative maximum point* (respectively *relative minimum point*) of the graph of f ;
- the number $f(x_0)$ is called a *relative maximum value* (respectively *relative minimum value*) of f .

Remark The terms *relative maximum value* and *relative minimum value* will not be used in this course because they are ambiguous; a value can be a relative maximum value as well as a relative minimum value.

If a function f has a relative maximum (respectively relative minimum) at x_0 , then the graph of f has a peak (respectively a valley) at $(x_0, f(x_0))$, that is, the point $(x_0, f(x_0))$ is higher than (respectively lower than) its neighboring points. However, it may not be the highest point (respectively lowest point) on the whole graph. For this reason, we say that $(x_0, f(x_0))$ is a *local maximum point* (respectively *local minimum point*) of the graph.

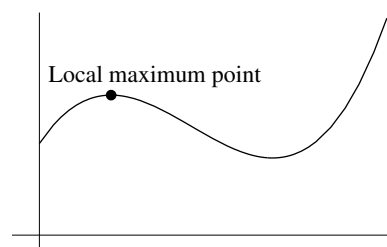


Figure 5.4

- The adjectives “*relative*” and “*local*” will be used interchangeably. For example, a local maximizer means a relative maximizer and a local minimum point means a relative minimum point etc.
- A local maximizer or a local minimizer will be called a *local extremizer* and a local maximum point or a local minimum point will be called a *local extremum point* etc.

Remark Consider the function $f : [0, 2] \rightarrow \mathbb{R}$ defined by

$$f(x) = -2x^2 + 3x + 1.$$

The graph of f is shown in Figure 5.5. Although $f(0) \leq f(x)$ for all x sufficiently close to and greater 0, the number 0 is not considered as a local minimizer of f .

- In order to consider whether a number is a local extremizer of a function, the function has to be defined on an open interval containing the number.

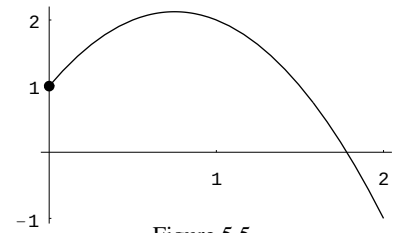


Figure 5.5

The following theorem gives a necessary condition for local extremizers.

Theorem 5.1.3 Let f be a function and let x_0 be a real number such that f is defined on an open interval containing x_0 . Suppose that f has a local extremum at x_0 . Then x_0 is a critical number of f , that is, $f'(x_0)$ does not exist or $f'(x_0) = 0$.

Proof Suppose that $f'(x_0)$ exists (that is, f is differentiable at x_0). We want to show that $f'(x_0) = 0$.

Without loss of generality, we may assume that f has a local maximum at x_0 (otherwise, we may consider the function $-f$ instead). By definition, there exists an open interval (a, b) with $x_0 \in (a, b) \subseteq \text{dom}(f)$ such that

$$f(x) \leq f(x_0) \quad \text{for all } x \in (a, b). \quad (5.1.1)$$

Note that $f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$; the limit exists and is a two-sided limit. Thus we have

$$f'(x_0) = \lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h}.$$

To show that $f'(x_0) = 0$, we show that one of the one-sided limits is at least zero and the other is at most zero.

- For $h < 0$ such that $a < x_0 + h$, by (5.1.1), we have $f(x_0 + h) \leq f(x_0)$ which implies that

$$\frac{f(x_0 + h) - f(x_0)}{h} \geq 0.$$

Hence we have $\lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} \geq 0$.

- For $h > 0$ such that $x_0 + h < b$, by (5.1.1), we have $f(x_0 + h) \leq f(x_0)$ which implies that

$$\frac{f(x_0 + h) - f(x_0)}{h} \leq 0.$$

Hence we have $\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} \leq 0$.

Therefore, the required result follows. $\square$

Remark For differentiable functions f , the result means that if f has a local extremum at x_0 , then $f'(x_0) = 0$.

Example Let f be the function given by $f(x) = 2\sqrt{x} - x$. We want to apply Theorem 5.1.3 to look for all the possible local extremizers of f .

Note that the domain of f is $[0, \infty)$. Thus local extremizers of f must belong to $(0, \infty)$. For $x > 0$, by the Power Rule, we have $f'(x) = \frac{1}{\sqrt{x}} - 1$. Thus f is differentiable on $(0, \infty)$. To look for local extremizers of f , by Theorem 5.1.3, we only need to find positive real numbers x_0 such that $f'(x_0) = 0$. Solving $\frac{1}{\sqrt{x}} - 1 = 0$, we get $x = 1$, which is the only possible candidate for local extremizer of f .

The next example shows that the converse Theorem 5.1.3 is not true: if $f'(x_0) = 0$, f may not have a local extremum at x_0 .

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x^3 - 3x^2 + 3x.$$

$$\begin{aligned} \text{Then we have } f'(x) &= \frac{d}{dx}(x^3 - 3x^2 + 3x) \\ &= 3x^2 - 6x + 3 \\ &= 3(x - 1)^2. \end{aligned}$$

Thus f is differentiable on $\mathbb{R}$. The number 1 is a critical number of f . However, it is not a local extremizer of f since f is increasing on $\mathbb{R}$.

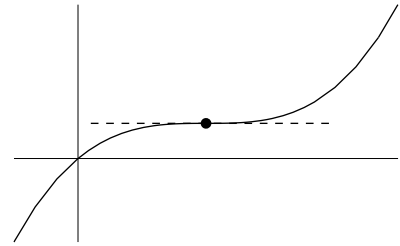


Figure 5.6

Suppose x_0 is a critical number of a function f . At x_0 , the function f may have a local maximum, a local minimum or neither. The next result describes a simple way to determine which case it is using the first derivative of f .

First Derivative Test Let f be a function that is differentiable on an open interval (a, b) and let $x_0 \in (a, b)$. Suppose that x_0 is a critical number of f .

- (1) If $f'(x)$ changes from positive to negative as x increases through x_0 , then x_0 is a local maximizer of f .
- (2) If $f'(x)$ changes from negative to positive as x increases through x_0 , then x_0 is a local minimizer of f .
- (3) If $f'(x)$ does not change sign as x increases through x_0 , then x_0 is neither a local maximizer nor local minimizer of f .

Explanation The assumption on x_0 is that $f'(x_0) = 0$.

- The condition “ $f'(x)$ changes from positive to negative as x increases through x_0 ” means that $f'(x) > 0$ for x sufficiently close to and less than x_0 and $f'(x) < 0$ for x sufficiently close to and greater than x_0 .
- The condition “ $f'(x)$ does not change sign as x increases through x_0 ” means that $f'(x)$ is either always positive or always negative for x sufficiently close and different from x_0 .

Proof We give the proof for (1) and (3). The proof for (2) is similar to that for (1).

- (1) If $f'(x)$ changes from positive to negative as x increases through x_0 , then there is an open interval in the form (a, x_0) such that $f'(x) > 0$ for all $x \in (a, x_0)$ and there is an open interval in the form (x_0, b) such that $f'(x) < 0$ for all $x \in (x_0, b)$; hence by Theorem 5.1.1 and continuity, f is increasing on $(a, x_0]$ and decreasing on $[x_0, b)$. Therefore, f has a local maximum at x_0 .
- (3) If $f'(x)$ does not change sign as x increases through x_0 , then there are open intervals in the form (a, x_0) and (x_0, b) such that $f'(x) > 0$ for all $x \in (a, x_0) \cup (x_0, b)$ or $f'(x) < 0$ for all $x \in (a, x_0) \cup (x_0, b)$. In the first case, by Theorem 5.1.1 f is increasing on (a, x_0) as well as on (x_0, b) and hence by continuity, it is increasing on (a, b) . In the second case, f is decreasing on (a, b) . Therefore, in any case, f does not have a local extremum at x_0 . $\square$

Remark For “nice” functions (for example, polynomials), the above result includes all possibilities. But we can construct weird functions f such that x_0 is a critical number of f and that f' changes sign infinitely often on the left and right of x_0 . Figure 5.7 shows the graph of the function f given below; the number 0 is a critical number of f .

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

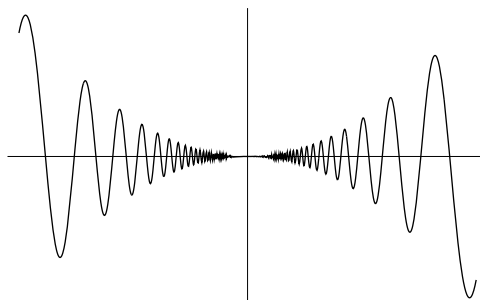


Figure 5.7

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = 27x - x^3.$$

Find and determine the nature of the critical number(s) of f .

Explanation The question is to find all the critical numbers of f and for each critical number, determine whether it is a local maximizer, a local minimizer or not a local extremizer.

Solution Differentiating $f(x)$, we get

$$\begin{aligned} f'(x) &= \frac{d}{dx}(27x - x^3) \\ &= 27 - 3x^2 \\ &= 3(3 + x)(3 - x). \end{aligned}$$

Solving $f'(x) = 0$, we get the critical numbers of f : $x_1 = -3$ and $x_2 = 3$.

- When x is sufficiently close to and less than -3 , $f'(x)$ is negative; when x is sufficiently close to and greater than -3 , $f'(x)$ is positive. Hence, by the First Derivative Test, $x_1 = -3$ is a local minimizer of f .
- When x is sufficiently close to and less than 3 , $f'(x)$ is positive; when x is sufficiently close to and greater than 3 , $f'(x)$ is negative. Hence, by the First Derivative Test, $x_2 = 3$ is a local maximizer of f .

□

Remark The function in the above example is considered in the last subsection in which a table is obtained. It is clear from the table that f has a local minimum at -3 and a local maximum at 3 . In the next example, we will use the table method to determine nature of critical numbers.

	$(-\infty, -3)$	$(-3, 3)$	$(3, \infty)$
3	+	+	+
$3 + x$	−	+	+
$3 - x$	+	+	−
f'	−	+	−
f	↘	↗	↘

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x^4 - 4x^3 + 5.$$

Find and determine the nature of the critical number(s) of the f .

Explanation The function is considered in an example in the last subsection. Below we just copy the main steps from the solution there.

Solution

$$\begin{aligned} f'(x) &= 4x^3 - 12x^2 \\ &= 4x^2(x - 3) \end{aligned}$$

	$(-\infty, 0)$	$(0, 3)$	$(3, \infty)$
f'	−	−	+
f	↘	↘	↗

Solving $f'(x) = 0$, we get the critical numbers of f : $x_1 = 0$ and $x_2 = 3$.

From the table, we see that

- the critical number $x_1 = 0$ is not a local extremizer of f ;
- the critical number $x_2 = 3$ is a local minimizer of f .

□

5.1.3 Convexity

In studying curves, we are also interested in finding out how the curves bend. Both curves shown in Figures 5.8(a) and (b) go up (as x goes from left to right). However the way how they bend are quite different.

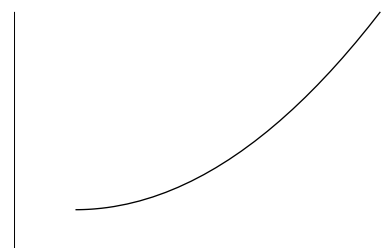


Figure 5.8(a)

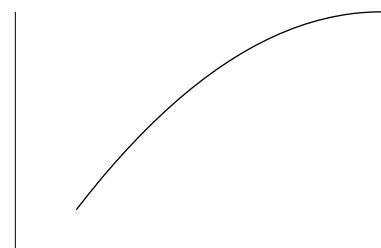


Figure 5.8(b)

- In Figure 5.8(a), the curve goes up faster and faster, that is, the slope becomes more and more positive as we move from left to right (the slope is increasing). We say that the curve is *bending up*.
- In Figure 5.8(b), although the curve goes up, the slope becomes less and less positive (the slope is decreasing). We say that the curve is *bending down*.

Similarly we can consider curves that go down.

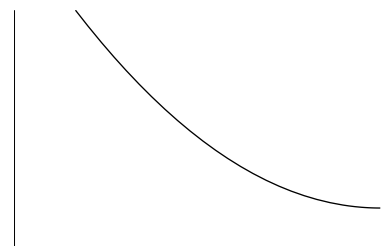


Figure 5.9(a)

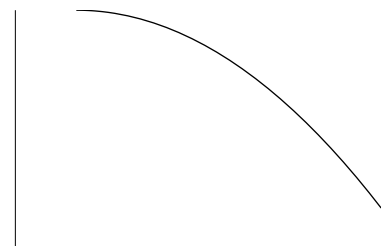


Figure 5.9(b)

- The curve in Figure 5.9(a) goes down. However, the slope becomes less and less negative. This means that the slope is increasing and we say that the curve is *bending up*.
- The curve in Figure 5.9(b) also goes down. Moreover, the slope becomes more and more negative. This means that the slope is decreasing and we say that the curve is *bending down*.

In summary, curves having shape shown in Figure 5.10(a) [or part of it] is said to be *bending up* and those having shape shown in Figure 5.10(b) [or part of it] is said to be *bending down*.

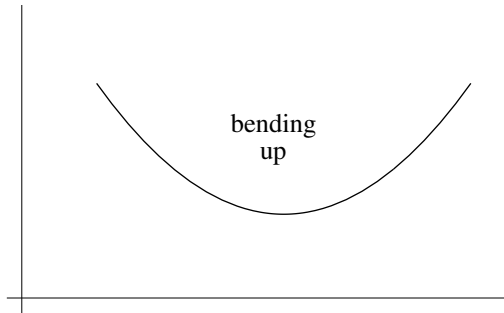


Figure 5.10(a)

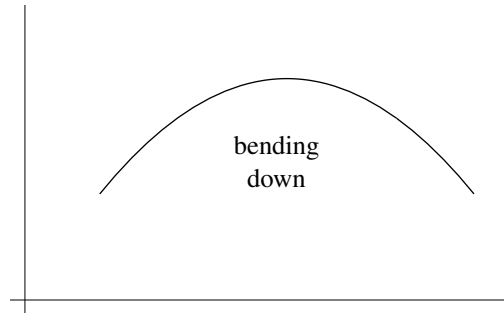


Figure 5.10(b)

Alternatively, if the curve is the graph of $y = f(x)$ where f is a differentiable function, the graph is bending up (respectively bending down) means that the graph is always above (respectively always below) the tangent lines.

Remark In many books, instead of bending up and bending down, the terms *concave up* and *concave down* respectively are used.

Bending up and bending down are properties of curves. Below are properties of functions corresponding to these geometric properties.

Definition Let f be a function that is defined and differentiable on an open interval (a, b) . We say that

- f is *strictly convex* on (a, b) if f' is increasing on (a, b) ;
- f is *strictly concave* on (a, b) if f' is decreasing on (a, b) .

Since f' is the slope function, f is strictly convex on (a, b) means that the slope is increasing and so in the interval (a, b) , the graph of f is bending up. Similarly, f is strictly concave means that in (a, b) , its graph is bending down.

Terminology For simplicity, instead of saying “strictly convex”, we will say “convex” etc.

The next theorem describes a simple way to find where a function is convex or concave. The method is to consider the sign of f'' .

Theorem 5.1.4 Let f be a function that is defined and is twice differentiable on an open interval (a, b) .

- (1) If $f''(x) > 0$ for all $x \in (a, b)$, then f is convex on (a, b) .
- (2) If $f''(x) < 0$ for all $x \in (a, b)$, then f is concave on (a, b) .

Proof We give the proof for (1). The proof of (2) is similar to that for (1).

Note that f'' is the derivative of f' . If $f''(x) > 0$ for all $x \in I$, that is, $(f')'(x) > 0$ for all $x \in (a, b)$, then by Theorem 5.1.1, f' is increasing on (a, b) , that is, f is convex on (a, b) . $\square$

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = 27x - x^3.$$

Find the interval(s) on which f is convex or concave.

Explanation

- The question is to find maximal open interval(s), if any, on which f is convex or concave.
- The given function f is a “nice” function (a polynomial function). It can be differentiated any number of times: $f^{(n)}(x)$ exists for all positive integers n and for all real numbers x . In particular, f is twice differentiable on $\mathbb{R}$. To apply Theorem 5.1.4, we have to solve inequalities $f''(x) > 0$ and $f''(x) < 0$. This is done by setting up a table.

Solution Differentiating $f(x)$, we get
$$\begin{aligned} f'(x) &= \frac{d}{dx}(27x - x^3) \\ &= 27 - 3x^2. \end{aligned}$$

Differentiating $f'(x)$, we get
$$\begin{aligned} f''(x) &= \frac{d}{dx}(27 - 3x^2) \\ &= -6x. \end{aligned}$$

	$(-\infty, 0)$	$(0, \infty)$
-6	$-$	$-$
x	$-$	$+$
f''	$+$	$-$

- On the interval $(-\infty, 0)$, f is convex.
- On the interval $(0, \infty)$, f is concave.

□

Remark

- When we consider “a function is convex/concave on an interval”, unlike increasing/decreasing, we do not include the endpoint(s) of the interval. This is because the concept is defined for open intervals only. Note that for a function f whose domain is a closed and bounded interval $[a, b]$, $f'(x)$ is undefined when $x = a$ or b .
- There is a more general definition for convex/concave functions. The definition does not involve f' and it can be applied to closed intervals also.

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x^4 - 4x^3 + 5.$$

Find where the graph of f is bending up or bending down.

Explanation This question is similar to the last one. The graph of f is bending up (or down) means that f is convex (or concave). So we have to find intervals on which f'' is positive (or negative).

Solution Differentiating $f(x)$, we get
$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^4 - 4x^3 + 5) \\ &= 4x^3 - 12x^2. \end{aligned}$$

Differentiating $f'(x)$, we get
$$\begin{aligned} f''(x) &= \frac{d}{dx}(4x^3 - 12x^2) \\ &= 12x^2 - 24x \\ &= 12x(x - 2). \end{aligned}$$

	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
$12x$	$-$	$+$	$+$
$x - 2$	$-$	$-$	$+$
f''	$+$	$-$	$+$

- On the intervals $(-\infty, 0)$ and $(2, \infty)$, the graph of f is bending up.

- On the interval $(0, 2)$, the graph of f is bending down.

□

Definition Let f be a function and let x_0 be a real number such that f is continuous at x_0 and differentiable on both sides of x_0 . If f is convex on one side of x_0 and concave on the other side, then we say that x_0 is an *inflection number* of f .

Explanation

- The condition “ f is differentiable on both sides of x_0 ” means that there is an open interval in the form (a, x_0) and an open interval in the form (x_0, b) such that $f'(x)$ exists for all $x \in (a, x_0) \cup (x_0, b)$.
- The condition “ f is convex on one side of x_0 and concave on the other side” means that there is an open interval in the form (α, x_0) and an open interval in the form (x_0, β) on which f is convex on one of them and concave on the other, that is, there is a change of convexity at x_0 .

Suppose that x_0 is an inflection number of a function f . By definition, on one side of the point $(x_0, f(x_0))$, the graph of f is bending up and on the other side, the graph is bending down. That is, there is a change of bending at the point $(x_0, f(x_0))$.

Terminology Suppose that x_0 is an inflection number of a function f . Then the point $(x_0, f(x_0))$ is called an *inflection point* of the graph of f .

In the following example, the function f is discussed in a previous example. Below we just copy part of the table obtained in the solution there.

Example Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = x^4 - 4x^3 + 5.$$

From the table

	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
f''	+	−	+

we see that the inflection numbers of f are 0 and 2.

Remark We can also say that the inflection points of the graph are $(0, 5)$ and $(2, -11)$.

The next result gives a necessary condition for inflection number.

Theorem 5.1.5 Let f be a function and let x_0 be a real number such that f is differentiable on an open interval containing x_0 and that $f''(x_0)$ exists. Suppose that x_0 is an inflection number of f . Then we have $f''(x_0) = 0$.

Proof By symmetry, we may assume that f is convex on the left-side of x_0 and concave on the right-side of x_0 , that is, there exist real numbers a and b with $a < x_0 < b$ such that f' is increasing on (a, x_0) and decreasing on (x_0, b) , which by continuity of f' at x_0 , implies that

$$f'(x) < f'(x_0) \quad \text{for all } x \in (a, x_0) \cup (x_0, b).$$

Thus, the function f' has a local maximum at x_0 . Hence by Theorem 5.1.3 (and using the assumption that the derivative of f' at x_0 exists), the derivative of f' at x_0 is 0, that is, $f''(x_0) = 0$. $\square$

Remark The converse of Theorem 5.1.5 is not true: if $f''(x_0) = 0$, x_0 may not be an inflection number of f .

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x^4.$$

Then we have $f'(x) = 4x^3$
 $f''(x) = 12x^2$

	$(-\infty, 0)$	$(0, \infty)$
f''	+	+

Although $f''(0) = 0$, the number 0 is not an inflection number of f . This is because f is convex on $(-\infty, 0)$ as well as on $(0, \infty)$.

Remark The function f is convex on $(-\infty, \infty)$.

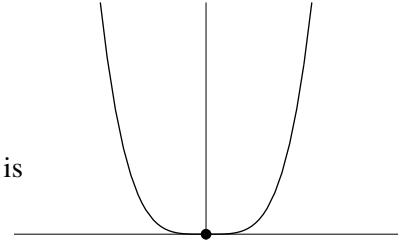


Figure 5.11

Terminology

- If $f'(x_0) = 0$, we say that x_0 is a stationary number of f . However, if $f''(x_0) = 0$, we do not have a specific name for x_0 .
- For local extremizers, there are two types: local maximizers and local minimizers. Correspondingly, there are also two types of inflection numbers. However, we do not have specific names to distinguish the two types.

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x^4 - 6x^2 + 5x - 6.$$

Find the inflection point(s) of the graph of f .

Explanation To find the inflection points of the graph, first we find the inflection numbers of the function. For that, we solve the equation $f''(x) = 0$. By Theorem 5.1.5, solutions to this equation include all the possible candidates for inflection numbers. However, for each of these candidates, we have to check whether the convexity of f are different on the left-side and right-side of it.

Solution Differentiating $f(x)$, we get $f'(x) = \frac{d}{dx}(x^4 - 6x^2 + 5x - 6)$
 $= 4x^3 - 12x + 5.$

Differentiating $f'(x)$, we get $f''(x) = \frac{d}{dx}(4x^3 - 12x + 5)$
 $= 12x^2 - 12$
 $= 12(x + 1)(x - 1).$

Solving $f''(x) = 0$, we get two solutions: $x_1 = 1$ and $x_2 = -1$.

	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
12	+	+	+
$x + 1$	−	+	+
$x - 1$	−	−	+
f''	+	−	+

From the table, we see that f is convex on $(-\infty, -1)$, concave on $(-1, 1)$ and convex on $(1, \infty)$. Hence $x_1 = 1$ and $x_2 = -1$ are the inflection numbers of f .

The inflection points of the graph are $(1, f(1)) = (1, -6)$ and $(-1, f(-1)) = (-1, -16)$. $\square$

To determine the nature of critical numbers, we can use the First Derivative Test discussed in the last subsection. Below, we discuss an alternative way using second derivatives.

Second Derivative Test Let f be a function and let x_0 be a real number such that f is differentiable on an open interval containing x_0 . Suppose that x_0 is a critical number of f , that is, $f'(x_0) = 0$.

- (1) If $f''(x_0) < 0$, then x_0 is a local maximizer of f (in fact, we have $f(x_0) > f(x)$ for all x sufficiently close to and different from x_0).
- (2) If $f''(x_0) > 0$, then x_0 is a local minimizer of f (in fact, we have $f(x_0) < f(x)$ for all x sufficiently close to and different from x_0).

Explanation Below we give a proof for (1). To prove (2), we can use the method for (1). Alternatively, we can apply (1) to the function $-f$ because $(-f)''(x_0) < 0$ in this case.

Proof It suffices to prove (1). Suppose that $f''(x_0) < 0$. We want to show that f is increasing on the left-side of x_0 and decreasing on the right-side.

By definition, together with the condition $f'(x_0) = 0$, we have

$$0 > f''(x_0) = \lim_{h \rightarrow 0} \frac{f'(x_0 + h)}{h},$$

which implies that $f'(x_0 + h) > 0$ if h is sufficiently close to and less than 0, that is,

$$f'(x) > 0 \quad \text{if } x = x_0 + h \text{ is sufficiently close to and less than } x_0.$$

Hence, by Theorem 5.1.1, f is increasing on the left-side of x_0 . Similarly, f is decreasing on the right-side of x_0 . Therefore, by the continuity of f at x_0 , we see that $f(x_0) > f(x)$ for all x sufficiently close to and different from x_0 . $\square$

Remark

- To determine the nature of a critical number using the Second Derivative Test, we consider the sign of f'' at the critical number. If we apply the First Derivative Test, we consider the sign of f' on the left-side and the right-side of the critical number.
- If $f''(x_0) = 0$, we can't apply the Second Derivative Test. At x_0 , the function f may have a local maximum, a local minimum or neither. See the last example in this subsection.

- Some students have the following conjecture:

Suppose $f'(x_0) = 0$ and x_0 is not a local extremizer of f , then x_0 is an inflection number of f .

For “nice” functions (for example, polynomial functions), the conjecture is correct. However, we can construct weird functions with “weird” critical point (see Figure 5.7).

However, when we consider nature of a critical number, there is no need to discuss whether it is an inflection number because critical numbers are related to first derivatives whereas inflection numbers are related to second derivatives.

Below, we redo a previous example using the Second Derivative Test.

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = 27x - x^3.$$

Find and determine the nature of the critical number(s) of f .

$$\begin{aligned} \text{Solution} \quad \text{Differentiating } f(x), \text{ we get } f'(x) &= \frac{d}{dx}(27x - x^3) \\ &= 27 - 3x^2 \\ &= 3(3 + x)(3 - x). \end{aligned}$$

Solving $f'(x) = 0$, we get the critical numbers of f : $x_1 = -3$ and $x_2 = 3$.

$$\begin{aligned} \text{Differentiating } f'(x), \text{ we get } f''(x) &= \frac{d}{dx}(27 - 3x^2) \\ &= -6x \end{aligned}$$

- At $x_1 = -3$, we have $f''(-3) = 18 > 0$; therefore, x_1 is a local minimizer of f .
- At $x_2 = 3$, we have $f''(3) = -18 < 0$; therefore, x_2 is a local maximizer of f .

□

Example Let f , g and h be functions from $\mathbb{R}$ to $\mathbb{R}$ given by

$$f(x) = x^4, \quad g(x) = -x^4, \quad h(x) = x^3.$$

It is clear that $x_1 = 0$ is a critical number of f , g and h . Moreover, we have $f''(0) = g''(0) = h''(0)$. However,

- at $x_1 = 0$, f has a local minimum;
- at $x_1 = 0$, g has a local maximum;
- at $x_1 = 0$, h does not have a local extremum.

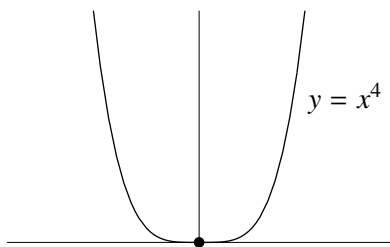


Figure 5.12(a)

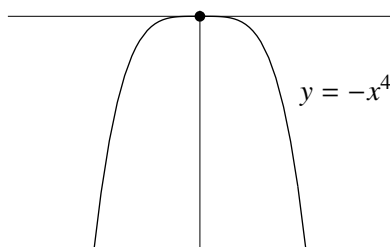


Figure 5.12(b)

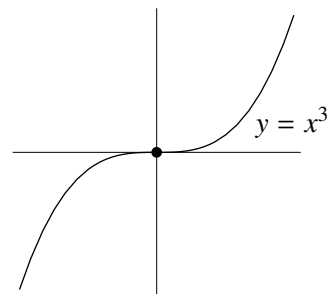
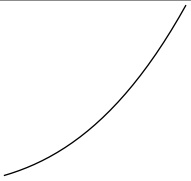
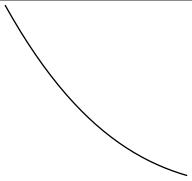
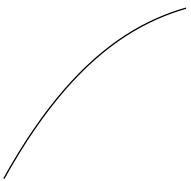
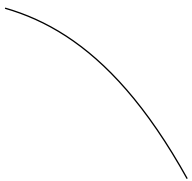


Figure 5.12(c)

5.1.4 Curve Sketching

Given a function f that is twice differentiable on an open interval (a, b) , to sketch the graph of $y = f(x)$ for $a < x < b$, we can use the first derivative of f to find where the graph goes up or down and use the second derivative of f to find where the graph bends up or down. Hence we can locate the local extremum points and inflection points of the graph. Intercepts give additional information for the graph. If f is a rational function, limits at infinity ($\pm\infty$) and vertical asymptotes (infinite limits) are also useful.

The following table gives the shape of the graph of f corresponding to the four cases determined by the signs of f' and f'' . For example, first row first column corresponds to that both f' and f'' are positive: the figure indicates that the graph goes up and bends up.

	$f' > 0$	$f' < 0$
$f'' > 0$		
$f'' < 0$		

Example Sketch the graph of $y = 27x - x^3$ for $x \in [-5.5, 5.5]$.

Explanation In the question, we are asked to draw the graph of f where $f(x) = 27x - x^3$ for $-5.5 \leq x \leq 5.5$. In the graph, we should locate the endpoints $(-5.5, f(-5.5))$ and $(5.5, f(5.5))$. In two previous examples, we obtain the following:

	$(-\infty, -3)$	$(-3, 3)$	$(3, \infty)$
f'	−	+	−

	$(-\infty, 0)$	$(0, \infty)$
f''	+	−

The three numbers -3 , 3 (zeros of f') and 0 (zero of f'') divide $\mathbb{R}$ into four intervals: $(-\infty, -3)$, $(-3, 0)$, $(0, 3)$ and $(3, \infty)$. On each of these intervals, we can use the above tables to consider the signs of f' and f'' .

Solution

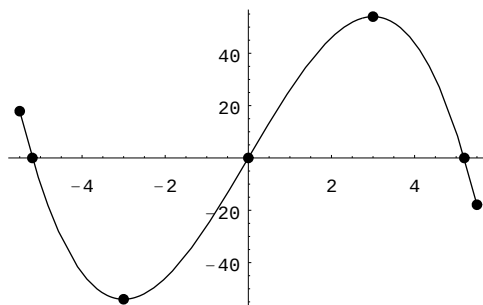
	$(-\infty, -3)$	$(-3, 0)$	$(0, 3)$	$(3, \infty)$
f'	−	+	+	−
f''	+	+	−	−
f				

On the graph, we have

- Local minimum point $(-3, f(-3)) = (-3, -54)$
- Inflection point $(0, f(0)) = (0, 0)$

- Local maximum point $(3, f(3)) = (3, 54)$
- Intercepts $(0, 0)$, $(3\sqrt{3}, 0)$ and $(-3\sqrt{3}, 0)$
- Endpoints $(-5.5, f(-5.5)) = (-5.5, 17.875)$
and $(5.5, f(5.5)) = (5.5, -17.875)$

The required graph is shown in the following figure:



□

Remark Since f is an odd function, the graph is symmetric about the origin.

Example Sketch the graph of $y = x^4 - 4x^3 + 5$ for $-1.5 \leq x \leq 4.2$.

Explanation In two previous examples, we obtain the following:

	$(-\infty, 0)$	$(0, 3)$	$(3, \infty)$
f'	−	−	+

	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
f''	+	−	+

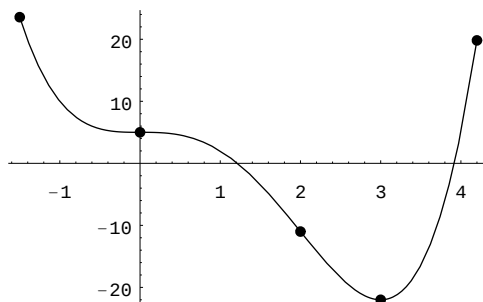
Solution

	$(-\infty, 0)$	$(0, 2)$	$(2, 3)$	$(3, \infty)$
f'	−	−	−	+
f''	+	−	+	+
f	\	\	\	/

On the graph, we have

- Inflection points $(0, f(0)) = (0, 5)$ and $(2, f(2)) = (2, -11)$
- Local minimum point $(3, f(3)) = (3, -22)$
- Endpoints $(-1.5, f(-1.5)) \approx (-1.5, 23.6)$
and $(4.2, f(4.2)) \approx (4.2, 19.8)$

The required graph is shown in the following figure:



□

Remark There are two x -intercepts. Approximate values of their x -coordinates are 1.2 and 3.9 which can be estimated using the Intermediate Value Theorem.

Example Sketch the graph of $y = x^3 + 3x^2 - 45x$ for $-9 \leq x \leq 6$.

Solution Differentiating $f(x)$, we get $f'(x) = 3x^2 + 6x - 45$
 $= 3(x-3)(x+5)$

	$(-\infty, -5)$	$(-5, 3)$	$(3, \infty)$
3	+	+	+
$x-3$	-	-	+
$x+5$	-	+	+
f'	+	-	+

Differentiating $f'(x)$, we get $f''(x) = 6x + 6$

	$(-\infty, -1)$	$(-1, \infty)$
f''	-	+

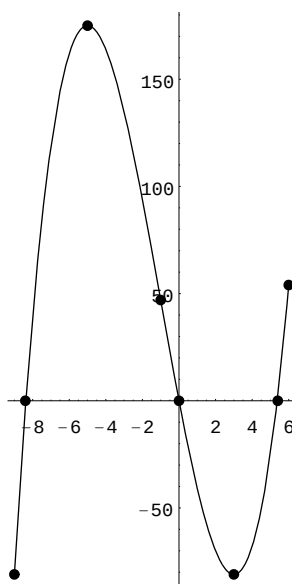
Combining the two tables, we get

	$(-\infty, -5)$	$(-5, -1)$	$(-1, 3)$	$(3, \infty)$
f'	+	-	-	+
f''	-	-	+	+
f	$\nearrow$	$\searrow$	$\searrow$	$\nearrow$

On the graph, we have

- Local maximum point $(-5, f(-5)) = (-5, 175)$
- Inflection point $(-1, f(-1)) = (-1, 47)$
- Local minimum point $(3, f(3)) = (3, -81)$
- Intercepts $(0, 0)$, $(\frac{-3+3\sqrt{21}}{2}, 0)$ and $(\frac{-3-3\sqrt{21}}{2}, 0)$
- Endpoints $(-9, f(-9)) = (-9, -81)$ and $(6, f(6)) = (6, 54)$

The required graph is shown in the following figure:



Exercise 5.1

- For each of the following functions f , find the interval(s) on which it is increasing.
 - $f(x) = 2x^2 - 5x + 6$
 - $f(x) = 1 + 3x - x^3$
 - $f(x) = x^3 + 6x^2 - 63x$
 - $f(x) = 2x^3 + 9x^2 - 6x + 7$
 - $f(x) = 3x^4 + 4x^3 - 24x^2 - 48x$
 - $f(x) = x + \frac{4}{x}$
- For each of the following functions f , find and determine the nature of its critical number(s).
 - $f(x) = -x^2 + 7x - 13$
 - $f(x) = x^4 - 2x^3$
 - $f(x) = x^5 - 15x^3$
 - $f(x) = \frac{x^2 + x + 1}{x + 1}$
- For each of the following functions f , find the interval(s) on which it is convex.
 - $f(x) = \sqrt{x}$
 - $f(x) = x^3 - 6x^2 + 9x$
 - $f(x) = 3x^5 - 9x^4 + 8x^3$
 - $f(x) = x + \frac{2}{x}$
- For each of the following functions f , find its inflection number(s).
 - $f(x) = 2x^3 + 9x^2 - 108x + 35$
 - $f(x) = 1 - \frac{1}{x} + \frac{1}{x^2}$
- For each of the following equations, sketch its graph (*you have to choose a suitable interval*).
 - $y = x^3 - 6x^2$
 - $y = 8x^3 - 2x^4$
 - $y = (x^2 - 3)^2$
 - $y = x^3 + x + 1$

5.2 Applied Extremum Problems

In this section, we will consider application of differentiation to applied extremum problems. In such problems, we are interested in *absolute* (or *global*) extrema rather than *relative* extrema.

5.2.1 Absolute Extrema

Definition Let f be a function and let x_0 be a real number belonging to the domain of f .

- If $f(x_0) \geq f(x)$ for all $x \in \text{dom}(f)$, then we say that f attains its (*absolute* or *global*) *maximum* at x_0 and that the number $f(x_0)$ is the (*absolute* or *global*) *maximum (value)* of f .
- If $f(x_0) \leq f(x)$ for all $x \in \text{dom}(f)$, then we say that f attains its (*absolute* or *global*) *minimum* at x_0 and that the number $f(x_0)$ is the (*absolute* or *global*) *minimum (value)* of f .

Remark Maximum and minimum values are unique (if exist).

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = x^2 + 1$. Then

- f attains its (absolute) minimum at 0 and the minimum of f is 1;
- f does not attain its (absolute) maximum, that is, there does not exist any $x_0 \in \mathbb{R}$ such that $f(x_0) \geq f(x)$ for all $x \in \mathbb{R}$.

Terminology Maximum and minimum (values) of a function f are called (*absolute* or *global*) *extrema* of f .

Recall (Extreme Value Theorem) Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then f attains its (absolute) maximum and minimum. That is, there exist $x_1, x_2 \in [a, b]$ such that

$$f(x_1) \leq f(x) \leq f(x_2) \quad \text{for all } x \in [a, b].$$

Note Extrema may occur at the endpoints a, b or at points in (a, b) .

Figure 5.13 shows the graph of a function f with domain $[a, b]$. Note that f attains its absolute minimum at x_2 which belongs to the open interval (a, b) and attains its absolute maximum at b which is an endpoint. Also note that f has a relative maximum at x_1 but it does not attain its absolute maximum there.

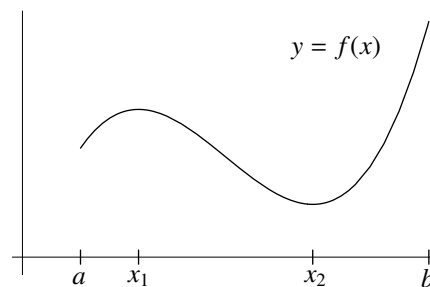


Figure 5.13

Let $f : [a, b] \rightarrow \mathbb{R}$ be a function that is differentiable on (a, b) . Suppose that f attains its maximum or minimum at x_0 where $a < x_0 < b$. Then by Theorem 5.1.3, x_0 must be a critical number of f , that is, $f'(x_0) = 0$. Thus we have the following procedures to find the absolute extrema of f .

Steps to find absolute extrema

- (1) Find the critical number(s) of f in (a, b) .
- (2) Find the values of f at the endpoints a and b and that at the critical number(s) found in (1).
- (3) The maximum and minimum values of f are, respectively, the greatest and smallest of the values found in Step 2.

FAQ Do we need to check the nature (relative maximum or minimum) of the critical numbers?

Answer If you want to find absolute extrema, there is no need to check the nature of the critical numbers. Even if you know that f has a local maximum (say) at a certain critical number x_0 , you still have to compare values. However if you know that f is increasing on $[a, x_0]$ and decreasing on $[x_0, b]$, then you can tell that f attains its absolute maximum at x_0 , that is, $f(x_0)$ is the absolute maximum; and to get the absolute minimum, you can compare the values $f(a)$ and $f(b)$. $\square$

Example Find the absolute extremum values of the function f given by

$$f(x) = 2x^3 - 18x^2 + 30x$$

on the closed interval $[0, 3]$.

Explanation In this question, the domain of f is taken to be $[0, 3]$. Since f is continuous on $[0, 3]$, it follows from the Extreme Value Theorem that f attains its absolute extrema. Note that f is differentiable on $(0, 3)$. Thus we can apply the above steps to find the absolute extremum values.

Solution Differentiating $f(x)$, we get

$$\begin{aligned} f'(x) &= \frac{d}{dx}(2x^3 - 18x^2 + 30x) \\ &= 6x^2 - 36x + 30 \quad (0 < x < 3) \end{aligned}$$

Solving $f'(x) = 0$, that is, $6x^2 - 36x + 30 = 0$ ($0 < x < 3$)

$$6(x-1)(x-5) = 0 \quad (0 < x < 3)$$

we get the critical number of f in $(0, 3)$: $x_1 = 1$.

Comparing the values of f at the critical number and that at the endpoints:

x	0	1	3
$f(x)$	0	14	-18

we see that the maximum value of f is 14 and the minimum value of f is -18. □

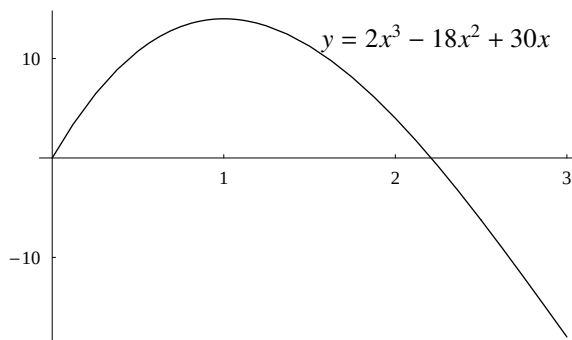


Figure 5.14

5.2.2 Applied Maxima and Minima

Example An article in a sociology journal stated that if a particular health-care program for the elderly were initiated, then t years after its start, n thousand elderly people would receive direct benefits, where

$$n(t) = \frac{t^3}{3} - 6t^2 + 32t, \quad 0 \leq t \leq 10.$$

After how many years does the number of people receiving benefits attain maximum?

Solution Differentiating $n(t)$, we get
$$\begin{aligned} n'(t) &= \frac{d}{dt} \left(\frac{t^3}{3} - 6t^2 + 32t \right) \\ &= t^2 - 12t + 32 \quad (0 < t < 10) \end{aligned}$$

Solving $n'(t) = 0$, that is, $t^2 - 12t + 32 = 0$ ($0 < t < 10$)

$$(t-8)(t-4) = 0 \quad (0 < t < 10),$$

we get the critical numbers of n in $(0, 10)$: $t_1 = 4$ and $t_2 = 8$.

Comparing the values of n at the critical numbers and that at the endpoints:

x	0	4	8	10
$n(x)$	0	$\frac{160}{3}$	$\frac{128}{3}$	$\frac{160}{3}$

we see that n attains its maximum at $t_1 = 4$ and also at $t_2 = 10$.

The number of people receiving benefits attains maximum after 4 years as well as after 10 years. □

Remark Although the maximum (if exist) of a function is unique, the above example shows that the values of x at which a function attains its maximum may not be unique. The following figure show the graph of the function n . Note that there are two highest points.

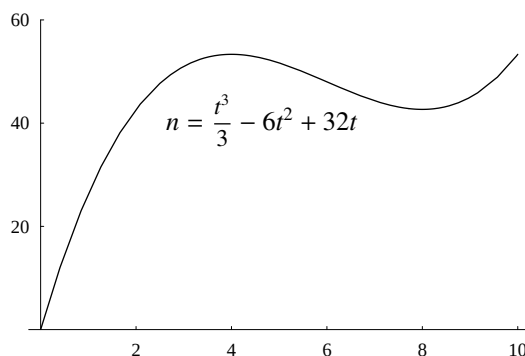


Figure 5.15

Example Find the dimensions of the rectangle that has maximum area if its perimeter is 20 cm.

Explanation The question asks for the length and width of the rectangle.

In the solution below, the domain of the area function A is not a closed interval. We can't use the steps as in the last example. Instead, we consider where A is increasing or decreasing.

Solution Let the length of one side of the rectangle be x cm.

Then the length of an adjacent side is $(10 - x)$ cm.

Note that $0 < x$ and $0 < 10 - x$. Thus we have $0 < x < 10$.

The area A (in cm^2) of the rectangle is

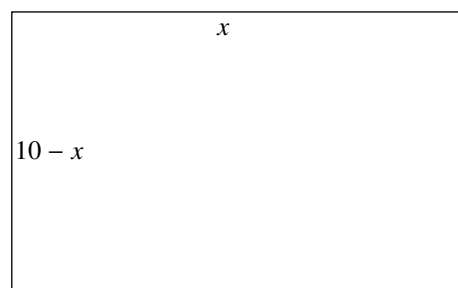


Figure 5.16

$$A(x) = x(10 - x), \quad 0 < x < 10.$$

We want to find the value of x at which A attains its maximum.

$$\begin{aligned} \text{Differentiating } A(x), \text{ we get } A'(x) &= \frac{d}{dx}(10x - x^2) \\ &= 10 - 2x \quad (0 < x < 10). \end{aligned}$$

Solving $A'(x) = 0$, we obtain the critical number of A : $x_1 = 5$.

Since A is increasing on $(0, 5)$ and decreasing on $(5, 10)$, it follows that A attains its absolute maximum at $x_1 = 5$.

	$(0, 5)$	$(5, 10)$
A'	+	-
A	$\nearrow$	$\searrow$

The dimensions of the largest rectangle is 5 cm $\times$ 5 cm.

□

Remark The largest rectangle is, in fact, a square.

FAQ Can we include 0 and 10 in the domain of A ?

Answer We may allow 0 and 10 in the domain of A . If $x = 0$ or 10, we get a rectangle one side of which is 0 cm. Such a figure is called a *degenerate rectangle*. Including the endpoints, the domain becomes a closed and bounded interval. Below we redo this problem using the method for the last example.

Alternative solution Let the length of one side of the rectangle be x cm. Then the length of an adjacent side is $(10 - x)$ cm. The area A (in cm^2) of the rectangle is

$$A(x) = x(10 - x), \quad 0 \leq x \leq 10.$$

Differentiating $A(x)$, we get $A'(x) = 10 - 2x$ ($0 < x < 10$).

Solving $A'(x) = 0$, we obtain the critical number of A : $x_1 = 5$. Comparing the values of A at the critical number and that at the endpoints:

x	0	5	10
$A(x)$	0	25	0

we see that A attains its maximum at $x_1 = 5$. Hence the dimensions of the largest rectangle is $5 \text{ cm} \times 5 \text{ cm}$. $\square$

FAQ Can we apply the Second Derivative Test to check that A has maximum at $x_1 = 5$?

Answer If you use the Second Derivative Test, you can only tell that A has *local* maximum at $x_1 = 5$. In this problem, we want *global* maximum.

However, there is a special version of the Second Derivative Test which can be applied to this problem. $\square$

Second Derivative Test (Special Version) Let f be a function and let x_0 be a real number such that f is differentiable on an open interval (a, b) containing x_0 . Suppose that x_0 is the only critical number of f in (a, b) .

- (1) If $f''(x_0) < 0$, then in (a, b) , f attains its maximum at x_0 , that is, $f(x_0) \geq f(x)$ for all $x \in (a, b)$.
- (2) If $f''(x_0) > 0$, then in (a, b) , f attains its minimum at x_0 , that is, $f(x_0) \leq f(x)$ for all $x \in (a, b)$.

Explanation Below we give a proof for (1). For this, we use a method called *Proof by Contradiction*. The result we want to prove is in the form “Assumption; Conclusion”.

- The assumption is “ f is differentiable on an open interval (a, b) containing x_0 and x_0 is the only critical number of f in (a, b) ”.
- The conclusion is “If $f''(x_0) < 0$, then in (a, b) , f attains its maximum at x_0 ”.

The negation (opposite) of the conclusion is “It is not true that if $f''(x_0) < 0$, then in (a, b) , f attains its maximum at x_0 ” which can be restated as “ $f''(x_0) < 0$ and in (a, b) , f does not attain its maximum at x_0 ”.

The method of *Proof by Contradiction* is to assume that the conclusion is false and use it (together with the given assumption) to deduce something that contradicts the given assumption. More specifically, we want to deduce that there exists $x_2 \in (a, b)$ with $x_2 \neq x_0$ such that $f'(x_2) = 0$, which contradicts the assumption that x_0 is the only critical number of f in (a, b) .

In the proof below, we write “Without loss of generality, we may assume that $x_1 > x_0$ ”. It means that the other case where $x_1 < x_0$ can be treated similarly.

Proof We give a proof for (1). For (2), it can be proved similarly or alternatively proved by applying (1) to the function $-f$.

Suppose that (1) does not hold, that is, suppose that $f''(x_0) < 0$ but there exists $x_1 \in (a, b)$ such that

$$f(x_1) > f(x_0).$$

Without loss of generality, we may assume that $x_1 > x_0$. Applying the Extreme Value Theorem to f on the interval $[x_0, x_1]$, we see that there exists $x_2 \in [x_0, x_1]$ such that

$$f(x_2) \leq f(x) \quad \text{for all } x \in [x_0, x_1].$$

It is clear that $x_2 \neq x_1$. Moreover, we have $x_2 \neq x_0$; this is because

$$f(x_0) > f(x) \quad \text{if } x \text{ is sufficiently close to } x_0 \text{ and } x \neq x_0$$

by the Second Derivative Test (since $f''(x_0) < 0$). Thus we have

$$x_2 \in (x_0, x_1) \quad \text{and} \quad f(x_2) \leq f(x) \text{ for all } x \in (x_0, x_1),$$

which implies that f has a local minimum at x_2 . By Theorem 5.1.3, we have $f'(x_2) = 0$, which (together with that $x_2 \neq x_0$) contradicts the assumption that x_0 is the only critical number of f in (a, b) . $\square$

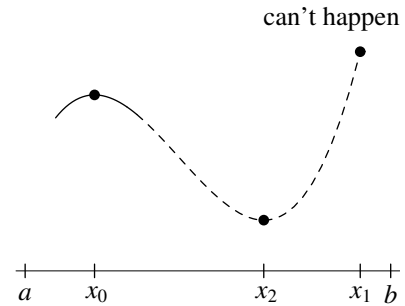


Figure 5.17

Alternative solution to the rectangle problem Let the length of one side of the rectangle be x cm. Then the length of an adjacent side is $(10 - x)$ cm. The area A (in cm^2) of the rectangle is

$$A(x) = x(10 - x), \quad 0 < x < 10.$$

Differentiating $A(x)$, we get $A'(x) = 10 - 2x$ ($0 < x < 10$).

Solving $A'(x) = 0$, we obtain the critical number of A : $x_1 = 5$.

Differentiating $A'(x)$, we get $A''(x) = \frac{d}{dx}(10 - 2x) = -2$.

Since $A''(5) = -2 < 0$ and 5 is the only critical number of A in $(0, 10)$, it follows from the Second Derivative Test (Special Version) that A attains its maximum at $x_1 = 5$. Hence the dimensions of the largest rectangle is $5 \text{ cm} \times 5 \text{ cm}$. $\square$

Example A rectangular box without lid is to be made from a square cardboard of sides 18 cm by cutting equal squares from each corner and then folding up the sides. Find the length of the side of the square that must be cut off if the volume of the box is to be maximized. What is the maximum volume?

Solution 1 Let the length of the side of the square to be cut off be x cm. Then the base of the box is a square with each side equals to $(18 - 2x)$ cm. Hence we have $0 < x < 9$.

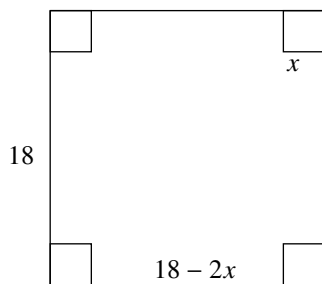


Figure 5.18

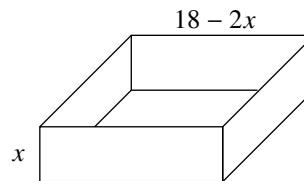


Figure 5.18(b)

The volume V , in cm^3 , of the open box is

$$V(x) = x(18 - 2x)^2, \quad 0 < x < 9.$$

$$\begin{aligned} \text{Differentiating } V(x), \text{ we get } V'(x) &= \frac{d}{dx}(324x - 72x^2 + 4x^3) \\ &= 324 - 144x + 12x^2 \quad (0 < x < 9) \end{aligned}$$

$$\begin{aligned}\text{Solving } V'(x) = 0, \text{ that is } \quad 324 - 144x + 12x^2 &= 0 \quad (0 < x < 9) \\ 12(x - 9)(x - 3) &= 0 \quad (0 < x < 9)\end{aligned}$$

we get the critical number of V in $(0, 9)$: $x_1 = 3$.

	$(0, 3)$	$(3, 9)$
V'	+	-
V	$\nearrow$	$\searrow$

Since V is increasing on $(0, 3)$ and decreasing on $(3, 9)$, it follows that on $(0, 9)$, V attains its maximum at $x_1 = 3$.

To maximize the volume of the box, the length of the side of the square that must be cut off is 3 cm.

The maximum volume is $V(3) = 432 \text{ cm}^3$. □

Solution 2 Let the length of the side of the square to be cut off be x cm. Then the base of the box is a square with each side equals to $(18 - 2x)$ cm. Hence we have $0 \leq x \leq 9$ (when $x = 0$ or 9 , we get a degenerate box with zero volume).

The volume V , in cm^3 , of the open box is

$$V(x) = x(18 - 2x)^2, \quad 0 \leq x \leq 9.$$

$$\begin{aligned}\text{Differentiating } V(x), \text{ we get } \quad V'(x) &= \frac{d}{dx}(324x - 72x^2 + 4x^3) \\ &= 324 - 144x + 12x^2 \quad (0 < x < 9)\end{aligned}$$

$$\begin{aligned}\text{Solving } V'(x) = 0, \text{ that is } \quad 324 - 144x + 12x^2 &= 0 \quad (0 < x < 9) \\ 12(x - 9)(x - 3) &= 0 \quad (0 < x < 9)\end{aligned}$$

we get the critical number of V in $(0, 9)$: $x_1 = 3$.

Comparing the value of V at the critical number and that the the endpoints:

x	0	3	9
$V(x)$	0	432	0

we see that to have maximum volume, the length of the side of the square that must be cut off is 3 cm; and that the maximum volume is 432 cm^3 . □

Solution 3 Let the length of the side of the square to be cut off be x cm. Then the base of the box is a square with each side equals to $(18 - 2x)$ cm. Hence we have $0 < x < 9$.

The volume V , in cm^3 , of the open box is

$$V(x) = x(18 - 2x)^2, \quad 0 < x < 9.$$

$$\begin{aligned}\text{Differentiating } V(x), \text{ we get } \quad V'(x) &= \frac{d}{dx}(324x - 72x^2 + 4x^3) \\ &= 324 - 144x + 12x^2 \quad (0 < x < 9)\end{aligned}$$

$$\begin{aligned}\text{Solving } V'(x) = 0, \text{ that is } \quad 324 - 144x + 12x^2 &= 0 \quad (0 < x < 9) \\ 12(x - 9)(x - 3) &= 0 \quad (0 < x < 9)\end{aligned}$$

we get the critical number of V in $(0, 9)$: $x_1 = 3$.

$$\begin{aligned}\text{Differentiating } V'(x), \text{ we get } \quad V''(x) &= \frac{d}{dx}(324 - 144x + 12x^2) \\ &= -144 + 24x.\end{aligned}$$

Since $V'''(3) = -72 < 0$ and $x_1 = 3$ is the only critical number of V in $(0, 9)$, it follows from the Second Derivative Test (Special Version) that in $(0, 9)$, V attains its maximum at $x_1 = 3$. Thus the length of the side of the square that must be cut off is 3 cm and the maximum volume is $V(3) = 432 \text{ cm}^3$. $\square$

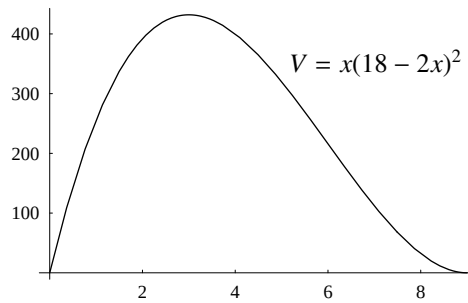


Figure 5.19

5.2.3 Applications to Economics

Suppose a manufacturer produces and sells a product. Denote $C(q)$ to be the total cost for producing and marketing q units of the product. Thus C is a function of q and it is called the *(total) cost function*. The rate of change of C with respect to q is called the *marginal cost*, that is,

$$\text{marginal cost} = \frac{dC}{dq}.$$

Denote $R(q)$ to be the total amount received for selling q units of the product. Thus R is a function of q and it is called the *revenue function*. The rate of change of R with respect to q is called the *marginal revenue*, that is,

$$\text{marginal revenue} = \frac{dR}{dq}.$$

Denote $P(q)$ to be the profit of producing and selling q units of the product, that is,

$$P(q) = R(q) - C(q).$$

Thus P is a function of q and it is called the *profit function*.

Denote $q_{\max}$ to be the largest number of units of the product that the manufacturer can produce. Assuming that q can take any value between 0 and $q_{\max}$. Then for each of the functions C , R and P , the domain is $[0, q_{\max}]$. Suppose that the cost function and the revenue function are differentiable on $(0, q_{\max})$ and suppose that producing 0 or $q_{\max}$ units of the product will not give maximum profit. Then in order to have maximum profit, we need

$$\frac{dP}{dq} = 0,$$

or equivalently,

$$\frac{dC}{dq} = \frac{dR}{dq},$$

that is, marginal cost = marginal revenue.

Example The demand equation for a certain product is

$$q - 90 + 2p = 0, \quad 0 \leq q \leq 90,$$

where q is the number of units and p is the price per unit, and the average cost function is

$$C_{\text{av}} = q^2 - 8q + 57 + \frac{2}{q} \quad 0 < q \leq 90.$$

At what value of q will there be maximum profit? What is the maximum profit?

Explanation Although the average cost function is undefined at $q = 0$, we may include 0 in the domain of the cost function. The cost function and the revenue function are differentiable on $(0, 90)$. However, we do not know whether maximum profit would be attained in $(0, 90)$ or at an endpoint. So we use the method for finding absolute extrema for functions on closed and bounded intervals.

Solution The cost function C is given by

$$C(q) = q \cdot C_{\text{av}} = q^3 - 8q^2 + 57q + 2 \quad (0 \leq q \leq 90),$$

and the revenue function R is given by

$$R(q) = p \cdot q = \frac{90-q}{2} \cdot q \quad (0 \leq q \leq 90).$$

Therefore the profit function P is given by

$$\begin{aligned} P(q) &= R(q) - C(q) \\ &= \left(45q - \frac{q^2}{2}\right) - (q^3 - 8q^2 + 57q + 2) \\ &= -q^3 + \frac{15}{2}q^2 - 12q - 2, \quad (0 \leq q \leq 90). \end{aligned}$$

$$\begin{aligned} \text{Differentiating } P(q), \text{ we get } P'(q) &= \frac{d}{dq} \left(-q^3 + \frac{15}{2}q^2 - 12q - 2 \right) \\ &= -3q^2 + 15q - 12 \quad (0 < q < 90) \end{aligned}$$

$$\begin{aligned} \text{Solving } P'(q) = 0, \text{ that is, } -3q^2 + 15q - 12 &= 0 \quad (0 < q < 90) \\ -3(q-1)(q-4) &= 0 \quad (0 < q < 90), \end{aligned}$$

we get the critical numbers of P : $q_1 = 1$ and $q_2 = 4$.

Comparing the values of P at the critical numbers as well as that at the endpoints:

q	0	1	4	90
$P(q)$	-2	$-\frac{15}{2}$	6	-669332

we see that maximum profit is attained at $q_2 = 4$ and the maximum profit is 6 (units of money). $\square$

Remark If we know that maximum profit is not attained at the endpoints, we can simply compare the values of P at q_1 and q_2 .

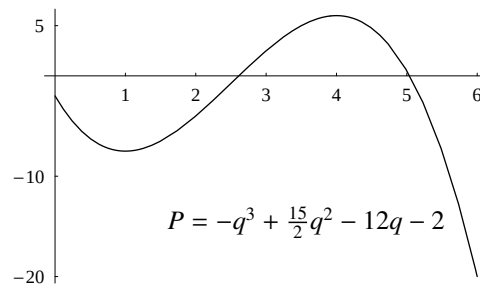


Figure 5.20

Exercise 5.2

- For each of the given function f , find its absolute extrema on the given interval.
 - $f(x) = 4x^3 + 3x^2 - 18x + 1$, $[0, 3]$
 - $f(x) = -3x^5 + 5x^3 + 2$, $[-2, 0]$
 - $f(x) = 1 + 2x^3 - 3x^4$, $[-1, 1]$
- Find two positive real numbers whose sum is 50 and whose product is a maximum.
- Find two real numbers x and y satisfying $2x + y = 15$ such that $x^2 + y^2$ is minimized.
Can you find a geometric meaning for the result?
- Find the dimensions of the rectangle of area 100 square units that has the least perimeter.
- A rectangular field is to be enclosed by a fence and divided equally into two parts by a fence parallel to one pair of the sides. If a total of 600 m of fence is to be used, find the dimensions of the field if its area is to be maximized.
- A book is to contain 36 in^2 of printed matter per page, with margins of 1 in along the sides and $1\frac{1}{2}$ in along the top and bottom. Find the dimensions of the page that will require the minimum amount of paper.
- Suppose that a ball is thrown straight up into the air and its height after t seconds is $5 + 24t - 16t^2$ feet. Determine how long it will take the ball to reach its maximum height and determine the maximum height.
- It is known from experiments that the height (in meter) of a certain plant after t months is given (approximately) by

$$h(t) = \sqrt{t} - t, \quad 0 \leq t \leq 1.$$

How long, on the average, will it take a plant to reach its maximum height? What is the maximum height?

- A company manufactures and sells x pieces of a certain product per month. The monthly cost (in dollars) is

$$C(x) = 120000 + 100x$$

and the price-demand equation is

$$p = 300 - \frac{x}{15}$$

where $0 \leq x \leq 4000$. Find the maximum profit, the production level that will give the maximum profit, and the price the company should charge for each piece of the product.